

09 • L4 — Executable Layer & UX

Assignment: “Implement the executable (end-user) layer (L4): Choose appropriate technology for implementation compatible with the L3 developed. Think about user interface considerations — what will make the system more or less acceptable and usable to end-users.”

9.1 Technology Decision

Option	Offline	Installation	Image UI freely designable	Cost	Effort	Decision
ODK Collect (Android)	✓	app store required	⚠ limited	free	low	✗
Enketo / web form	⚠ partly	none	⚠ limited	free	low	✗
Native app (iOS/Android)	✓	app store required	✓	free	high	✗
Self-contained web app (single HTML file, no framework)	✓	none	✓	free	medium	✓

Rationale

No installation. Persona 1 (58 years old, medium tech literacy, recently operated on) should be able to scan a QR code at hospital discharge and start using the app immediately. Every obstacle — app store, account, update — costs adherence, and adherence is the critical variable for a *daily* check.

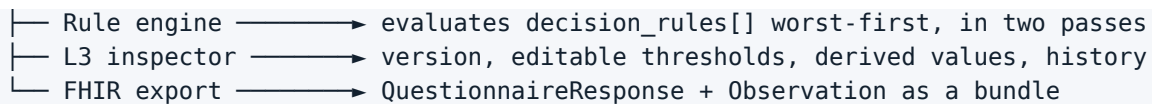
Full control over the image interaction. The large-format wound appearance cards are not decoration but the central clinical solution to the translation problem (C-01). With standard ODK widgets, the L4 would have dictated the UX — precisely the anti-pattern that Day 3 describes under “L4 constraints on L3”.

Offline as the default state, not as a feature. The app is a single HTML file with embedded React. After the first load it runs without any network connection at all. For home-based aftercare this is not an optimisation but a prerequisite (NFR-2).

The price: we forgo an existing ecosystem (server infrastructure, user management, synchronisation mechanisms of ODK Central). For a prototype this is acceptable; for a rollout a synchronisation layer would have to be added (11 Implementation).

9.2 How the L4 Executes the L3

```
wundcheck-app.html      (single file, ~56 KB, no dependencies, no build step)
├─ L3 loader ──────────> embedded L3, or any JSON via "Load L3 file"
├─ Renderer ───────────> builds the questions from elements[], grouped by sections[]
├─ i18n ───────────────> language switch from meta.languages, label[lang]
└─ Expression engine ──> evaluates relevant / calculation / constraint
```



The app contains not a single clinical statement. No threshold, no question and no recommendation is hard-coded. Using the **“Load L3 file”** (“L3-Datei laden”) button, any other L3 can be loaded — the app would then represent an entirely different CDSS.

Demo proof: change `meta.thresholds.fever_orange` from `38.0` to `37.5`, reload the app, enter the same case → the result switches from GREEN to ORANGE. Without a rebuild, without a developer, without a single line of code.

9.3 UX Decisions

Decision	Rationale	Persona
Image selection instead of description for wound appearance and discharge	Overcomes language and educational barriers; produces structured, comparable data instead of free text	Peter
One question per screen , large touch targets	Small screens, reading glasses, possibly impaired fine motor skills after surgery	Peter
Baseline data only on first use (gateway “First use?” / “Erste Nutzung?”)	Daily use must stay under 3 minutes (NFR-3), otherwise adherence collapses	Peter
Sequence follows patient logic , not guideline structure	First “how am I feeling”, then “what does the wound look like” — inspecting the wound requires opening the dressing and therefore comes last	Peter
Colour-coded traffic light + plain-language action	Colour alone is neither accessible (colour vision deficiency) nor action-guiding. Every colour is accompanied by a concrete sentence: “Contact your practice today ” (“Kontaktieren Sie heute Ihre Praxis”)	Peter
The result states the reason , not just the level	Counteracts automation bias: the patient should be able to follow the reasoning behind the recommendation rather than obey it blindly. “Because of the erythema extending beyond the wound edge...” (“Wegen der Rötung über den Wundrand hinaus...”)	Peter
Reassurance carries equal weight to warning	A CDSS that only warns forfeits half of its benefit (Scenario A)	Peter
Constraint messages in everyday language	“Plausible temperature: 34–43 °C” (“Plausible Temperatur: 34–43 °C”) instead of “Validation error: value out of range”	Peter
The alert contains context , not just a status	Sandra must be able to triage within 10 seconds: classification, triggering rule, vital signs, 7-day trend	Sandra
Alerts only for ORANGE/RED	An alert for every submission would cause alert fatigue in the practice — and then nothing at all gets read	Sandra
Version visible in the header	It remains traceable which rule version produced a given recommendation	Anna

What We Deliberately Did Not Build

Omitted	Why
Photo upload / image analysis	Image quality, lighting and skin type vary considerably; ML decisions require explanation; regulatory demands are considerably higher. Structured image selection delivers comparable data without these problems

Omitted	Why
User accounts / login	An obstacle at first use. For the prototype the data stay local; in a rollout a token-based assignment without a password would be conceivable
Push notifications	Requires a native app or a service worker plus server. In the prototype this is replaced by the reminder given during the discharge consultation
Multilingual support	Provided for in the L3 (<code>meta.language</code> , label structure), but only <code>de</code> is populated. Essential for a rollout, not for the prototype
A real HIS connection	The FHIR export produces a valid bundle as a download — the interface is demonstrated without a hospital information system being required

This is scope discipline, not a gap. The assignment explicitly places quality above completeness.

9.4 FHIR Interface

The interface is **not** a file download. The app builds an R4 **transaction bundle** and POSTs it to the endpoint configured in the L3.

Artefact	FHIR resource	Purpose	When
Form definition	<code>Questionnaire</code> (<i>conceptually — our L3</i>)	Structure, questions, visibility logic	—
Daily check	<code>QuestionnaireResponse</code>	All answers, typed (<code>valueInteger</code> , <code>valueDecimal</code> , <code>valueDate</code> , <code>valueCoding</code>), canonical reference carries the L3 version	every run
Classification	<code>Observation</code>	Traffic-light result, Southampton grade, triggering rule, L3 version, days post-op — displayable in the HIS as a trend curve without opening the individual answers	every run
Alert	<code>Communication</code>	Structured notification to the practice, <code>priority: urgent</code> (ORANGE) or <code>stat</code> (RED)	only ORANGE/RED

```
// meta.fhir in the L3 – configuration, not code
"fhir": {
  "version": "R4",
  "endpoint": "https://hapi.fhir.org/baseR4",          // point at the hospital server or the
  EPD gateway
  "questionnaire_canonical": "http://wundcheck.example/Questionnaire/wundcheck",
  "patient_reference": "Patient/example",
  "alert_recipient": "Organization/practice",
  "send_mode": "transaction"
}
```

The endpoint is a piece of L3 configuration. Redirecting WundCheck from the test server to a hospital FHIR server or the Swiss EPD gateway means editing one line of JSON — not a single line of app code changes. That is the same argument as for the thresholds, applied to the interface.

Offline first. *Send to patient record* posts the bundle. If the endpoint is unreachable, the bundle goes into an outbound queue, the queue length is visible in the L3 inspector, and *Retry queue* flushes it. The classification and the recommendation are shown to the patient regardless — transmission failure must never block care advice. *Download FHIR bundle* remains available as a fallback and for the demo.

Alerts only from ORANGE upwards. A `Communication` is generated only for ORANGE and RED. GREEN and YELLOW are written into the record but do not notify anyone — this is the alert-fatigue mitigation from 12, implemented in code rather than merely promised in prose.

Conformance-tested. `tools/check-fhir.mjs` extracts the bundle builder from the app and checks 20 R4 shape rules — see 10 §10.5. A sample bundle is in `submission/example-fhir-bundle.json`.

Why no framework at all

The first version used React with an esbuild bundle. We replaced it with plain JavaScript for three reasons:

1. **No build step.** A content change to the L3 was already rebuild-free — but changing the *rendering* required Node, npm and a bundler. The whole L4 is now one file that opens in any editor.
2. **Reviewability.** The application is roughly 480 lines of readable JavaScript. A maintainer can verify for themselves that it holds no clinical logic — which is exactly the claim we make in 08, and which `tools/check-hardcoding.mjs` enforces automatically.
3. **Longevity.** No dependency can go stale, become vulnerable or disappear. For a tool meant to be maintained by a clinical department rather than a software team, that outweighs developer convenience.

The price: no component ecosystem, everything hand-written. At this size that is a good trade.

Interoperability level achieved

Measured against the four levels from Lecture Day 4:

Level	Achieved?	Reasoning
1 Foundational	✓	Data exchange is technically possible — the app POSTs a bundle to a FHIR endpoint
2 Structural	✓	FHIR R4 resources with a defined structure; typed answers; versioned canonical
3 Semantic	⚠ partial	38 of 48 elements carry a terminology binding. Full semantics would require complete mapping and a resolved SNOMED CT licensing position
4 Organisational	✗	Governance, consent management, the legal framework and process integration are not addressed

The honest answer: **structurally yes, semantically partly, organisationally no.**

9.5 Open Points

- ☒ New elements and rules from 05/06 implemented in the L3 and tested (19/19)
- ☒ Outbound queue for pending FHIR transmissions implemented
- ☒ Application rebuilt without a framework — no build step, no dependencies
- ☐ Wound graphics are schematic placeholders; they should be redrawn against the Southampton grades and for more than one skin tone
- ☐ A graphic for `redness_extent` (5 levels) is still missing — the question is currently text-only
- ☐ Third language for the pilot population

Related documents: 08 L3 · 10 QA · 11 Implementation